Term 1 answer key

Formulas printed for students in every question. Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

WEEK	SESSION	Q1 A Angles & shapes	Q2 B Measurement	Q3 C Number & percentages	Q4 D Algebra, ratio & rates	Q5 E Number plane, data & chance
1	1	27°	50.3 m	−3	$4m + 36$	164
	2	116°	42 cm^3	6.4	$300 and $200	5/8
	3	75°	60 m^2	75%	$x = 24$	−4, 2
2	1	50°	52 cm	16	41	13
	2	110°	39 m^2	30 L	60 km/h	1/2
	3	37°	2800 m	9%	$2 per kg	9/25
3	1	143°	50 cm^2	12	$8m + 8n$	2
	2	71°	120 cm^2	95%	9:05 am	50
	3	60°	31.4 cm	10	$x = 8$	3/10
4	1	66°	64 cm^2	9	16	(3, 5)
	2	115°	48 m	$6	3 : 4	1/5
	3	150°	75 m^3	0.75	$45 and $75	9
5	1	110°	30 cm^2	0.45	$15x - 10$	9/10
	2	45°	58 cm^2	85%	2 m/s	2
	3	109°	42 cm^2	45	Bottle B ($3 per L)	3
6	1	58°	3000 cm^3	27 L	$y = 23$	7/9
	2	76°	44.0 cm	−2	8	1/2
	3	118°	30 cm	49	4 h 45 min	8, 12
7	1	42°	99 m^2	90%	$6a + 6$	23
	2	115°	56 cm^2	84%	8 : 3	30
	3	65°	96 cm^2	28 m	$210 and $150	4
8	1	72°	72 cm^3	$30	$p = 6$	3/10
	2	142°	60 cm^2	30.6	15	7/12
	3	56°	31.4 cm	3/5	90 km/h	1/6
9	1	38°	4 L	−1	$3x + 15$	(3, 5)
	2	123°	20 m	26%	Box B ($23 per kg)	28
	3	82°	400 m^2	$175	35 minutes	12
10	1	66°	90 cm^2	4	$m = 9$	14
	2	130°	52 m^2	$36	6	7/25
	3	140°	80 cm^2	16/25	2 : 5	5/12

A MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

B MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

C MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

D MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

E MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.